

Recall:

Prop 2.3.14. $f: X \rightarrow Y$ morph. of var. $\forall F \in D_c^b(Y, \mathbb{k}), f^*F \in D_c^b(X, \mathbb{k})$

2.3.15. $U \hookrightarrow X$ loc. closed subvar. $\forall F \in D_c^b(Y, \mathbb{k}), h_1 F \in D_c^b(X, \mathbb{k})$

2.3.17. $F, G \in D_{\text{qcf}}^b(X, \mathbb{k}), F \otimes G$ & $R\text{Hom}(F, G) \in D_{\text{qcf}}^b(X, \mathbb{k})$

2.3.18. $\in D_c^b \quad F \otimes G \in D_c^b$

2.3.19. f : finite morph. $F \in D_c^b \Rightarrow f_* F \in D_c^b$

(Artin vanishing) 2.4.2

X smooth var. Z div. with snc. $j_Z: X_Z \hookrightarrow X$ inclusion of stratum in the normal crossing strat Z is finite type on X_Z
 $\Rightarrow j_{Z*} \mathbb{Z}$ constr. & codim supp $H^i(j_{Z*} \mathbb{Z}) = |Z| + i$.

• Sheaf functor & constructibility

Prop: $f: X \rightarrow Y$ morph. of var. $F \in D_c^b(X, \mathbb{k})$

$f_* F \in D_c^b(Y, \mathbb{k})$

Remark: $f^!$ leave for Verdier Dual Class.

• Induction on $\dim X$

• Truncation

$\Rightarrow F$ constructible sheaf.

Negate: $X \xrightarrow{i} \bar{X} \xrightarrow{p} Y$

open: $f_* F \vee$

proper: $f_* F = f_* f^* F$

i open embedding
 p proper.

X irr.

$i: X_i \hookrightarrow X$ inclusion of an irr. comp.

$j_i: U_i \hookrightarrow X$ comp.

$f_* j_{i*} F \rightarrow f_* F \rightarrow (f_* i)_* i^* F$

$U_i \xrightarrow{j_i} X \xrightarrow{f} Y$
 $j_i \searrow \quad \quad \quad \nearrow f$
 $\quad \quad \quad U_i \xrightarrow{f_i} Y$

$f_* j_{i*} = f_* j_{i*}$ & Both f_* & $(f_* i)_*$ are of fewer irr. comp.

$P(U) \xrightarrow{\tilde{f}} \tilde{Y} \xleftarrow{\tilde{p}} \tilde{P}(\tilde{Z})$
 $P \downarrow \quad \quad \quad \downarrow \tilde{P}$
 $U \hookrightarrow X \hookrightarrow Y \xleftarrow{i} Z$

smooth open set

$\mathbb{L} := F|_U$ local system

• assume f dominant

$Z := Y \setminus U \Rightarrow$ proper $p: \tilde{Y} \rightarrow Y$

"Resolution of sing var. $P(U) \Rightarrow U$ "

2.1.18 $P(U)$ div. with simple n.c.

$f_* h_* \mathbb{L} \cong p_* h_* \mathbb{L}$

open $\Rightarrow (f_* h_* \mathbb{L})|_U$ constr.

enough to

$i^* f_* h_* \mathbb{L} \cong i^* p_* h_* \mathbb{L}$

$\cong p_* i^* h_* \mathbb{L}$

$\cong p'_* (U|_{\tilde{Z}} \mathbb{L}|_{\tilde{Z}})$

$\uparrow$

Constr. 2.4.2. good stratification

& p'_* : induction $\dim \tilde{Z}' < \dim U = \dim X$

$\Rightarrow h_* \mathbb{L}$ constr.

$(f_* i)_* i^* F \rightarrow f_* F \rightarrow f_* h_* \mathbb{L} \rightarrow$

Reduction:

$f_* h_* \mathbb{L}$ constr.

$U \hookrightarrow X \xrightarrow{j} W$

$\downarrow f$

Y

$i_* i^* F \rightarrow F \rightarrow h_* \mathbb{L}$

• F constr.

$i: Z \hookrightarrow X$

• $h_* \mathbb{L}$ constr. by Step 2 $\Rightarrow i_* i^* F$ constr.

$\Rightarrow i^* F$ constr.

$\dim Z < \dim X \Rightarrow (f_* i)_*$ preserves constr. by ind.

$\underbrace{(f_* i)_* i^* F}_{\uparrow} \rightarrow f_* F \rightarrow f_* h_* \mathbb{L} \rightarrow$

• preserves.

• to prove $f_* F$ constr. only need to prove $f_* h_* \mathbb{L}$ constr.

• Induction on $\dim Y$.

• $\dim Y = 0 \Rightarrow R\Gamma(\mathbb{L})$ constr. by Artin vanishing thm

• $\dim Y > 0$ f proper $p: \tilde{X} \rightarrow X$

$P(U) \xrightarrow{\tilde{f}} \tilde{X} \xleftarrow{\tilde{p}} \tilde{P}(\tilde{Z})$

$\parallel \quad \downarrow P \quad \downarrow \tilde{P}$

$U \hookrightarrow X \hookrightarrow Z$

$\downarrow f \quad \downarrow \tilde{f}$

Y

$f_* h_* \mathbb{L} \cong \tilde{f}_* h_* \mathbb{L}$

$\tilde{Z}_1 \dots \tilde{Z}_k$ irr comp of $\tilde{Z}$

2.1.13 $\Rightarrow \exists V \subset Y$ st.

$\exists \tilde{f}^{-1}(V) \cap \tilde{Z}_i: \tilde{f}^{-1}(V) \cap \tilde{Z}_i \rightarrow V$

smooth & proper

$\Rightarrow \tilde{f}^{-1}(V) \rightarrow V$ surj

• $\tilde{f}: V \rightarrow V$ transverse to $\tilde{f}^{-1}(U) \xrightarrow{\tilde{f}} \tilde{X} \xleftarrow{\tilde{p}} \tilde{P}(\tilde{Z})$

$\tilde{Z} \cap V$

$\downarrow \tilde{f}$

$\downarrow \tilde{f}$

$\downarrow \tilde{f}$

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Thm: 2.7.1. $f: X \rightarrow Y$ morph.

$F \in D_c^b(X, \mathbb{k}) \quad f_* F, f^* F \in D_c^b(Y, \mathbb{k})$

lem 2.7.2. $i: Z \hookrightarrow X$ closed emb.

$F \in D_c^b(X, \mathbb{k}) \Rightarrow i^* F \in D_c^b(Z, \mathbb{k})$

$j: U \hookrightarrow X \hookrightarrow Z$ open

$i_* i^* F \rightarrow F \rightarrow j_* j^* F \rightarrow$

F constr. $\vee \quad j^* F$ constr. $\wedge$ $j_* j^* F$ constr. 2.7.1

$\Rightarrow i_* i^* F$ constr.

$\Rightarrow i^! = i^* (i_* i^! F)$

Prop 2.7.3. $F, G \in D_c^b(X, \mathbb{k})$

$\Rightarrow R\text{Hom}(F, G) \in D_c^b(X, \mathbb{k})$

$U \hookrightarrow X \hookrightarrow Z$

$i_* R\text{Hom}(i^* F, i^* G) \rightarrow R\text{Hom}(F, G) \rightarrow j_* R\text{Hom}(F|_U, G|_U) \rightarrow$
 $\uparrow \quad \quad \quad \uparrow$
constr. constr.

Exer 1.3.7: like

$i_* i^* F \rightarrow F \rightarrow j_* j^* F$

& $R\text{Hom}(F|_U, G|_U)$ 2.3.17

$\Rightarrow \square$

Cohomology vanishing result.

Thm 2.7.4. X var. of $\dim n$ F constr. sheaf on X ,

$\Rightarrow H^k(X, F) = 0 \quad \forall k > 2n$ & vanishes unless $0 \leq k \leq 2n$

$\alpha_X: X \rightarrow \text{pt}$

$\alpha_X^* F$ constr.

$R\Gamma(X, j_* j^* F \rightarrow F \rightarrow i_* i^* F \rightarrow)$

$R\Gamma(X, F|_U) \rightarrow R\Gamma(X, F) \rightarrow R\Gamma(X, F|_Z) \rightarrow$

• $\dim Z \leq n$ induction $\Rightarrow H^k(Z, F|_Z) = 0$

$m > 2n \quad H^m(U, F|_U) = 0$ largest.

0: $R\Gamma(X, F|_U) \rightarrow T^{2m} R\Gamma(X, F|_U) \cong M[-m]$

$\text{Hom}(U, (F|_U) \otimes M[-m]) \cong$

$\text{Hom}(F|_U, \alpha_U^* M[-m])$

Recall 2.2.1

$\text{Hom}(F|_U, \alpha_U^* M[-m]) \cong$

$\cong \text{Ext}^{2n-m}(F|_U, M(n))$

But $2n-m < 0 \quad \text{Ext} \text{ vanishes}$

Thm 2.7.5. $H^k(X, F) = 0 \quad \forall k > 2n$ & vanishes unless $0 \leq k \leq 2n$

Can't use argument of 2.7.4 because

$(a_1, a_1) \vee$

$(a^* a_*) \times$

Alternative proof:

$\forall$ var. X of $\dim n$ admits a covering

by at most $n+1$ affine open subvar.

$\Rightarrow F$ constr.

$H^k(X, F) = 0 \quad \forall k > 2n$ & vanishes unless $0 \leq k \leq 2n$

Prop 1.3.13 LNV dist. t.v.)

Y_1, Y_2 both closed or open

$Y_1 \cup Y_2 = X$

$h_1: Y_1 \hookrightarrow X, h_2: Y_2 \hookrightarrow X$

$h: Y_1 \cap Y_2 \hookrightarrow X \quad \forall F \in D^+(X, \mathbb{k})$

$F \rightarrow h_{1*} h_1^* F \oplus h_{2*} h_2^* F \rightarrow h_* h^* F \rightarrow$

$h_* h^* F \rightarrow h_{1*} h_1^* F \oplus h_{2*} h_2^* F \rightarrow F \rightarrow$

(Unlabeled proof ref. Achar)

Step 1: Proven for all closed subvar of X

$G \in D_c^b(X, \mathbb{k})$ s.t. codim supp $H^i(G) \geq i$ for $i \leq n$

& $H^i(G) = 0$ if $i > n$

$\Rightarrow H^k(X, G) = 0 \quad \forall k > 2n$

induction on # nonzero coh.

$T \xrightarrow{\epsilon^{m-1}} G \rightarrow H^m(G)[-m] \rightarrow$ sheaves of G

$\Rightarrow R\Gamma(\tau^{\leq m} G) \rightarrow R\Gamma(G) \rightarrow R\Gamma(H^m(G)[-m])$

$H^k(X, \tau^{\leq m} G) = 0 \quad \forall k > 2n$ by ind

$H^k(X, H^m(G)[-m])$

$H^m(G)$ supported on a closed subvar of $\dim \leq m$

assumption $\Rightarrow H^k(X, H^m(G)[-m]) = 0$ if $k > 2n - 2m$

$\Rightarrow H^k(X, H^m(G)[-m]) = 0$ if $k > 2n - m$

2.7.5 = 0

Step 2. X proper: $R\Gamma(F) \cong R\Gamma(F|_U)$

Step 3. X smooth & F loc. sys

$j: X \hookrightarrow \bar{X}$ open dense.

2.1.18 assume $\bar{X}$ smooth. $\bar{X} \setminus X$ div. with simple n.c.

$j_* F$ satisfies 2.

Step 1. $H^k(X, F) \cong H^k(\bar{X}, j_* F) = 0 \quad \forall k > 2n$ Lemma 2.4.2.

Step 4: general: ind on $\dim X$

& use ind on vars

X pt: t.v.

$j: U \hookrightarrow X$ irr. smooth Zar. open

$i: Z \hookrightarrow X \quad j_* j^* F \rightarrow F \rightarrow i_* i^* F \rightarrow$

$\Rightarrow R\Gamma(j_* \mathbb{L}) \rightarrow R\Gamma(F) \rightarrow R\Gamma(i_* \mathbb{L})$

enough to $H^k(X, j_* \mathbb{L}) = 0 \quad \forall k > 2n$

$U \hookrightarrow \bar{U} \hookrightarrow X$, assume $X = \bar{U}$ X irr.

U irr. smooth Zar. open subset s.t. $F|_U$ is

j factors through

$\bar{U} \hookrightarrow X$. assume for simplicity $X = \bar{U} \Rightarrow X$ irr.

$U \hookrightarrow X \hookrightarrow Z$

$\parallel \quad \uparrow P \quad \uparrow \quad j^* \cong P \otimes \mathbb{Z}$

$P(U) \xrightarrow{\tilde{f}} \tilde{X} \xleftarrow{\tilde{p}} \tilde{P}(\tilde{Z}) \Rightarrow R\Gamma(j_* \mathbb{L}) \cong R\Gamma(\tilde{j}_* \mathbb{L})$ enough to prove $H^k(\tilde{X}, \tilde{j}_* \mathbb{L}) = 0$

smooth

$\tilde{j}_* \tilde{j}^* \mathbb{L} \rightarrow \tilde{j}_* \mathbb{L} \rightarrow \tilde{i}_* \tilde{i}^* \mathbb{L} \rightarrow$

$H^i(\tilde{j}_* \mathbb{L}) \cong H^i(\tilde{i}_* \tilde{i}^* \mathbb{L}) \quad \forall i \geq 1$

lem 2.4.2 $\Rightarrow$

codim supp $H^i(\tilde{i}_* \tilde{i}^* \mathbb{L}) \geq i$

$\forall i \geq 1$

$\Rightarrow \text{codim supp } H^i(\tilde{i}_* \tilde{i}^* j_* \mathbb{L}[i]) \geq i$

$\forall i \geq 0$

$\Rightarrow H^k(\tilde{i}_* \tilde{i}^* j_* \mathbb{L}[i]) = 0 \quad \forall k > 2(n-1)$

$\Rightarrow H^k(\tilde{i}_* \tilde{i}^* j_* \mathbb{L}[i-1]) = 0 \quad \forall k > 2n$

Use $R\Gamma \Rightarrow$

$R\Gamma(\tilde{i}_* \tilde{i}^* \mathbb{L}[i-1]) \rightarrow R\Gamma(j_* \mathbb{L}) \rightarrow R\Gamma(\mathbb{L}) \rightarrow$

Step 3 $\Rightarrow H^k(U, \mathbb{L}) = 0 \quad \forall k > 2n$

Upper $\Rightarrow H^k(\tilde{Z}, \tilde{i}^* j_* \mathbb{L}[i-1]) = 0 \quad \forall k > 2n$

$\Rightarrow H^k(X, j_* \mathbb{L}) = 0 \quad \forall k > 2n \quad \square$

$f: X \rightarrow Y$ smooth morph. of smooth var.
 $Z \subset X$ div. with snc. f transv. locally t.v. fib.
 $F \in D^+(X, \mathbb{k})$ weakly constr. w.r.t. n.c. str.
 $\Rightarrow f_* F \in D_{\text{loc}}^b(Y, \mathbb{k})$
moreover $(f_* F)_j \cong R\Gamma(F|_{g^{-1}(j)})$ isom. (Remark)
 f smooth $\Rightarrow \dim f^{-1}(y)$
 $= \dim X - \dim Y < \dim X$